snowflake

From Air Quality to Aircraft and Automobiles To Ghosts Data Is Everywhere

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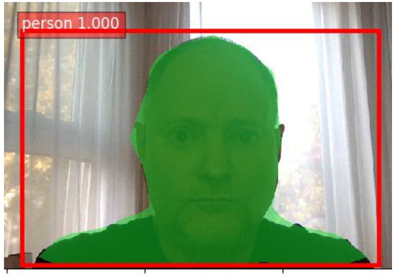
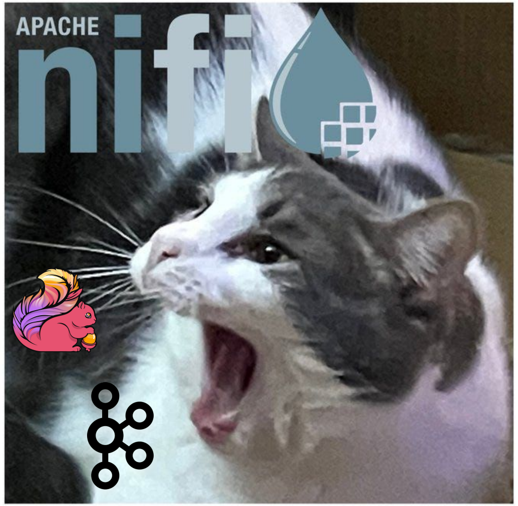


Streamlit

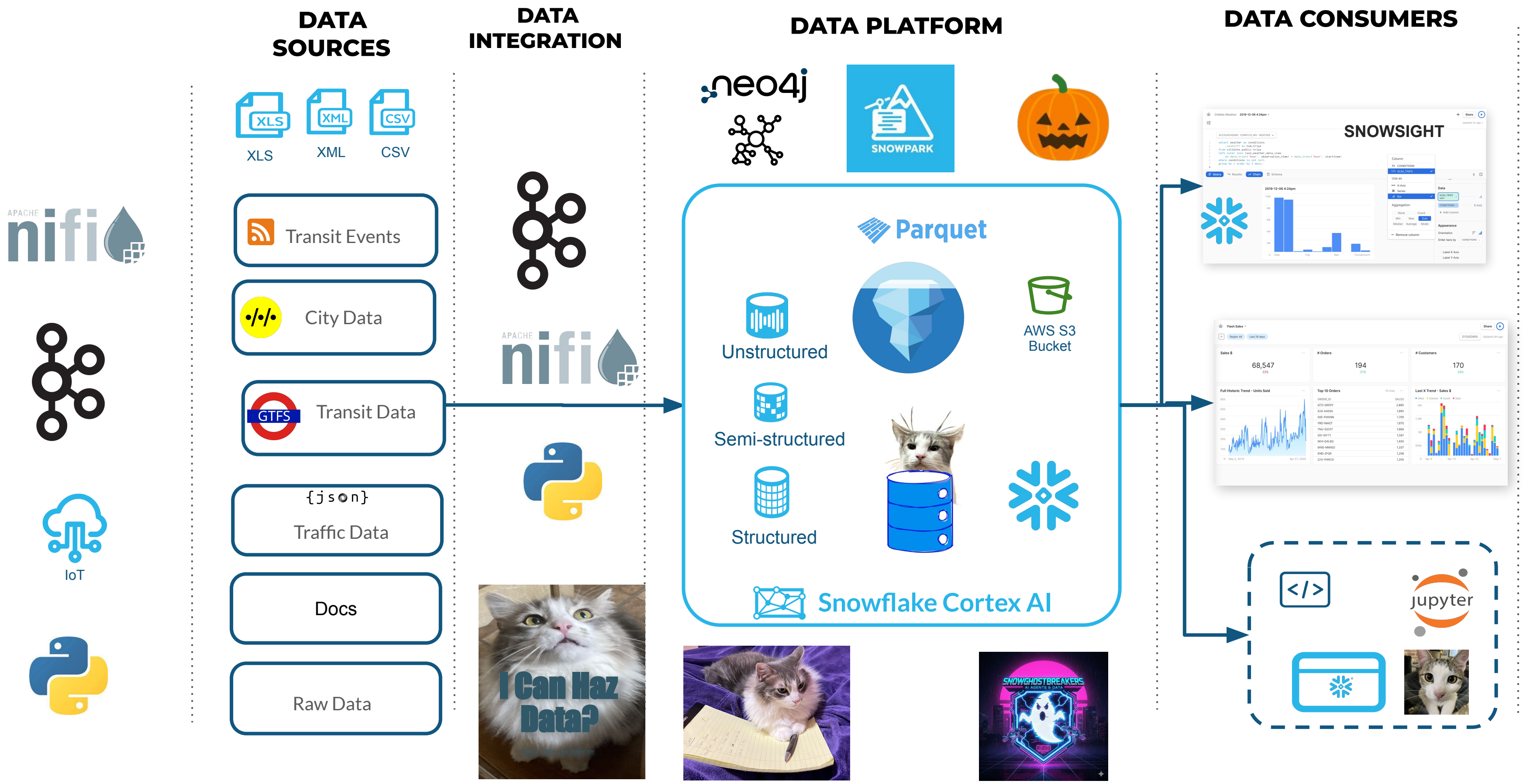
AI + Streaming Weekly by Tim Spann



https://bit.ly/32dAJft

This week in Snowflake, Apache NiFi, Apache Flink, Apache Kafka, ML, AI, Streamlit, Jupyter, Apache Iceberg, Apache Polaris, Python, Java, LLM, GenAI, Vectors and Open Source friends.

Real-Time AI Graph Enhanced Open Lakehouse



Graph Algorithms in Snowflake

Pathfinding & Search
(Find a path)

- A* Shortest Path
- All Pairs Shortest Path
- Breadth & Depth First Search
- Delta-Stepping Single-Source
- Dijkstra's Single-Source
- Dijkstra Source-Target
- Minimum Weight Spanning Tree
- Minimum Weight k-Spanning Tree
- Random Walk
- Yen's K Shortest Path
- Minimum Directed Steiner Tree
- Topological Sort
- Longest Path

Link Prediction
(Are things related)

- Adamic Adar
- Common Neighbors
- Preferential Attachment
- Resource Allocations
- Same Community
- Total Neighbors

Community Detection
(Group or Split)

- Weakly Connected Components
- Conductance Metric
- K-1 Coloring
- K-Means Clustering
- K-core Decomposition
- Label Propagation
- Leiden
- Local Clustering Coefficient
- Louvain
- Max K-Cut
- Modularity Optimization
- Speaker Listener Label Propagation
- Strongly Connected Components
- Triangle Count

Centrality
(What is important)

- ArticleRank
- Betweenness Centrality & Approx.
- Closeness Centrality
- Degree Centrality
- Eigenvector Centrality
- Harmonic Centrality
- Hyperlink Induced Topic Search (HITS)
- Influence Maximization (CELF)
- PageRank
- Personalized PageRank

Similarity
(Who are alike?)

- K-Nearest Neighbors (KNN)
- Node Similarity
- Filtered KNN & Node Similarity
- Cosine & Pearson Similarity Functions
- Euclidean Distance Similarity Function
- Euclidean Similarity Function
- Jaccard & Overlap Similarity Functions

Graph Embeddings
(Numerical representation of graph data)

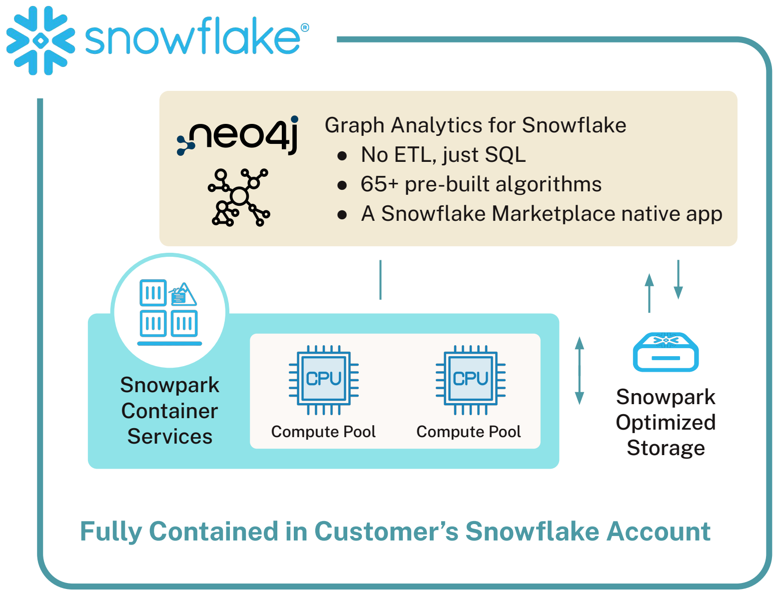
- Fast Path
- Fast Random Projection (FastRP)
- FastRP with Property Weights
- GraphSAGE
- Node2Vec
- HashGNN
- Knowledge Graph Embeddings
- Subgraph embeddings

...and more!

The Native App Keeps Data Secure and Compliant Within Snowflake

Run analytical jobs on demand using SQL. These:

1. Create a graph from Snowflake tables on Snowpark (SPCS)
2. Run the graph algorithms
3. Write the results back to Snowflake tables for downstream analytics or reporting.



Run on-demand graph analytics with SQL. Create in-memory graphs from Snowflake tables, apply algorithms, and write results back for downstream use–all fully contained in Snowflake.

Neo4j Graph Analytics for Snowflake

Seamlessly bring graph-powered insights to your Snowflake AI Data Cloud

Uncover Hidden Patterns in Your Snowflake Data

Enrich data for downstream analytics, visualization, and apps including Streamlit in Snowflake

No ETL, Just SQL

Eliminate costly ETL by running algorithms and writing results directly to Snowflake tables

Serverless, Elastic Analytics in Snowflake

Reduce administrative costs with a fully managed analytics offering that uses the built-in governance provided by Snowflake

visit the Neo4j Graph Analytics for Snowflake listing in Snowflake Marketplace.

Unstructured Data

Unstructured

- Lots of formats
- Text, Documents, PDF
- Images, Videos, Audio
- Email, Slack, Teams
- Logs
- Binary Data Formats
- Zip, Archives
- Variants

Semi-Structured Data

Semi-structured

- Open Data like Open AQ - Air Quality Data
- Location, Time,Sensors
- Apache Avro, Parquet, Orc
- JSON and XML
- Hierarchical Data
- Logs
- Key-Value

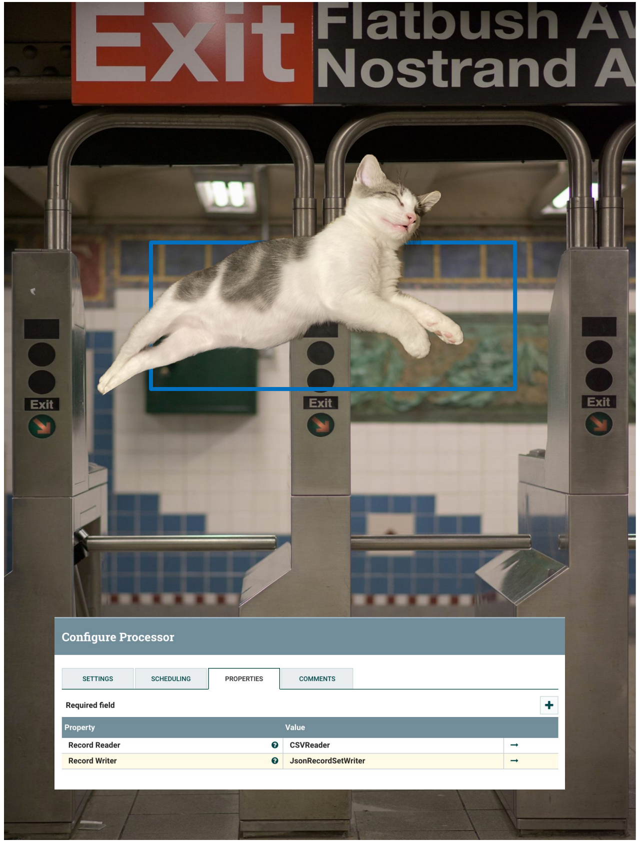
https://docs.snowflake.com/en/sql-reference/data-types-semistructured

Structured Data

Structured

- Snowflake Tables
- Snowflake Hybrid Tables
- Apache Iceberg Tables
- Relational Tables
- Postgresql Tables
- CSV, TSV

Apache NiFi



- From laptop to 1,000 nodes
- Ingest, Extract, Split
- Enrich, Transform
- Mature, 10 years+
- Any Data, Any Source
- LLM Calls
- Data Provenance
- Back Pressure
- Guaranteed Delivery

Apache NiFi for Data Ingest, Movement and Routing

- Guaranteed delivery
- Data buffering
  - Backpressure
  - Pressure release
- Prioritized queuing
- Flow specific QoS
- Data provenance
- Supports push and pull models

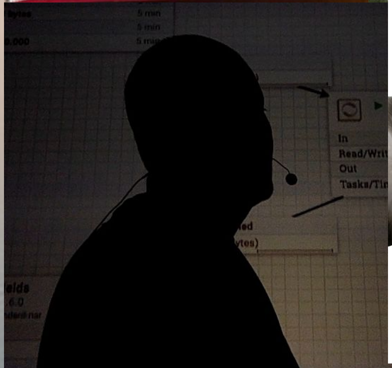
- Hundreds of processors
- Visual command and control
- Hundreds of sources
- Flow templates
- Pluggable/multi-role security
- Designed for extension
- Clustering
- Version Control

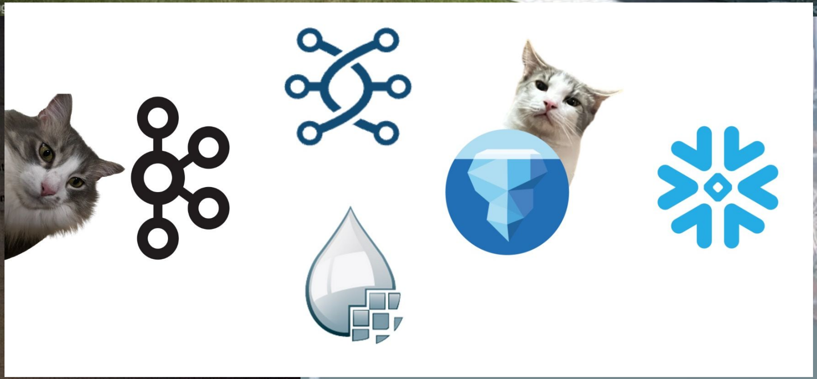
The Power of Apache NiFi

- Moving Binary, Unstructured, Image and Tabular Data
- REST, TCP/IP, UDP
- Enrichment
- Universal Visual Processor
- Simple Event Processor
- Routing
- Feeding data to Central Messaging
- Support for modern protocols
- Kafka Protocol Source/Sink
- Universal REST API


SPIRIT
Business
Cat
ADULT
Size Costume
ONE SIZE FITS MOST


AUTOMATE EVERYTHING





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